



Turning Agricultural waste to Carbon Capture Solutions

Background

India generates approximately 25 million tonnes of rice husk annually, of which only ~20% is productively utilized. The remaining husk is largely disposed/ openly burned, releasing ~22 million tonnes of CO₂ per year.

When combusted, rice husk produces ash containing 85–90% amorphous silica, presenting a significant yet underutilized opportunity.

Zeolites are critical materials in refining, power generation, chemical processing, and environmental remediation, but high costs limit their accessibility for Indian MSMEs.

Our Solution

- ✓ Convert rice husk waste into high-purity, silica-rich zeolites at commercial scale.
- ✓ Produce affordable zeolite products for CO₂ capture, flue gas treatment, and industrial pollutant adsorption.
- ✓ Enable small and medium enterprises in North-East India to adopt cleaner processes through cost-effective, locally produced materials.
- ✓ Establish decentralized production and supply chains to strengthen regional circular economy and reduce emissions.



Rice-husk-derived zeolites enable circular utilization of agricultural waste by acting as low-cost adsorbents for flue gas treatment, CO₂ capture, gas purification, wastewater treatment, and industrial emission control, thereby reducing pollution and improving air quality.

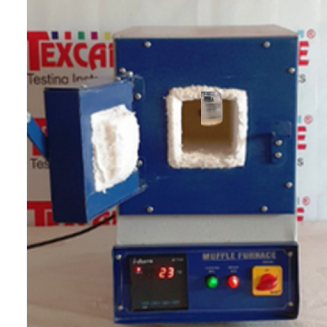
OUR TEAM

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Testing Phase



Rice husk ash and NaOH solution were mixed with sodium aluminate, stirred thoroughly, and left to age overnight to form a homogeneous precursor gel.



The mixture was heated in an oven at 85–90 °C for 6 hours, during which solid zeolite crystallized and settled at the bottom of the hydrothermal steel reactor.



The formed zeolite was filtered, washed, and dried in an oven at 100–105 °C, yielding dry zeolite powder.

Yield and carbon capture

Theoretical calculations indicate that 1 g of RHA can produce ~2 g of zeolite, while experimental synthesis achieved ~1.5 g of zeolite per gram of RHA (76.8% yield). Based on the measured adsorption capacity of the zeolite (~3.5 mmol CO₂/g), this corresponds to a carbon capture potential of about 0.23 g of CO₂ per gram of RHA at lab scale.

Zeolite Pellet Formation and Activation

- Zeolite powder is mixed with a small amount of binder and water, shaped into pellets by pressing or extrusion, and oven-dried to obtain mechanically stable adsorbent pellets.
- The dried pellets are thermally activated at 300–450 °C to remove moisture and open pore structures, enhancing CO₂ adsorption performance.

Comparison and economics

Rice husk-derived zeolite replaces costly commercial raw materials with agricultural waste, making production more economical. The process uses mild conditions and simple equipment, lowering energy and operational needs. Modular design enables decentralized production near raw-material sources. At ₹110–125 per kg, it costs nearly one-fourth of high-grade commercial zeolites (~₹400 per kg) while supporting sustainable carbon capture.

Improvisation

- ✓ Seed-assisted crystallization: Introduced 1–2% zeolite seed to accelerate nucleation, improve crystallinity, and enhance overall yield.
- ✓ Gel aging: We let the aluminosilicate gel formed from rice husk ash-derived silica solution and sodium aluminate solution to age before hydrothermal treatment, resulting in improved phase purity and crystal growth.

Implementation Roadmap

